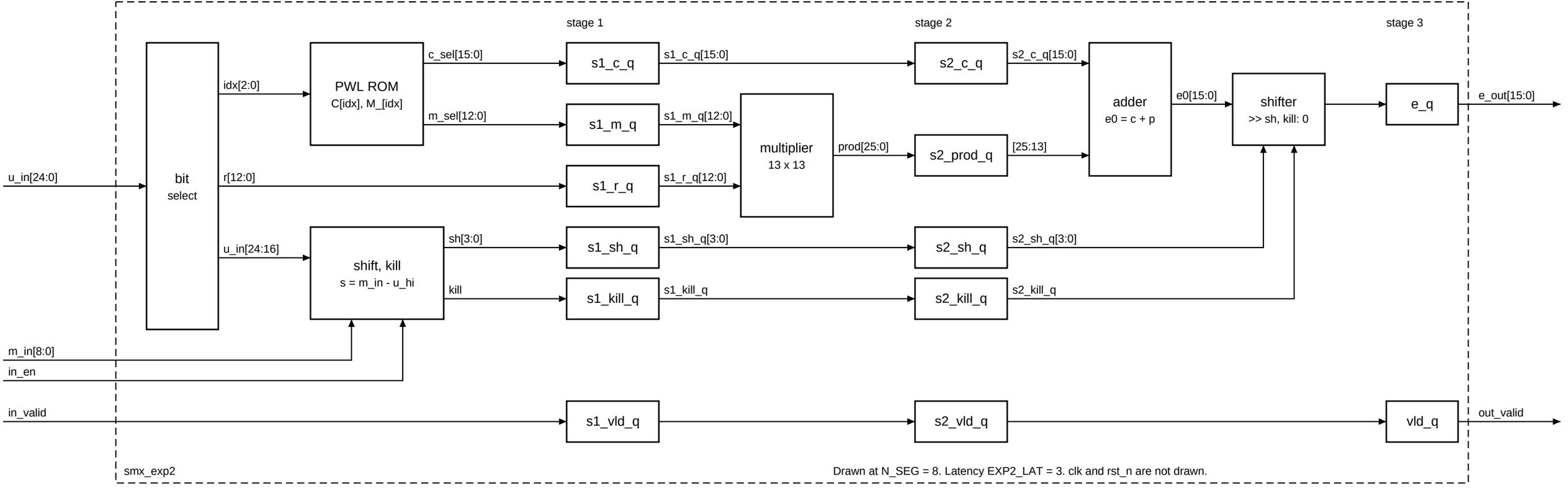
stage 1
stage 2
stage 3
u_in[24:0]
bit
select
idx[2:0]
r[12:0]
u_in[24:16]
PWL ROM
C[idx], M_[idx]
c_sel[15:0]
m_sel[12:0]
shift, kill
s = m_in - u_hi
sh[3:0]
kill
m_in[8:0]
in_en
in_valid
s1_c_q
s1_m_q
s1_r_q
s1_sh_q
s1_kill_q
s1_vld_q
s1_c_q[15:0]
s1_m_q[12:0]
s1_r_q[12:0]
s1_sh_q[3:0]
s1_kill_q
multiplier
13 x 13
prod[25:0]
s2_c_q
s2_prod_q
s2_sh_q
s2_kill_q
s2_vld_q
s2_c_q[15:0]
[25:13]
s2_sh_q[3:0]
s2_kill_q
adder
e0 = c + p
e0[15:0]
shifter
>> sh, kill: 0
e_q
vld_q
e_out[15:0]
out_valid
smx_exp2
Drawn at N_SEG = 8. Latency EXP2_LAT = 3. clk and rst_n are not drawn.